CENTRAL STORE MANAGEMENT SYSTEM V2

Production Deployment & Operations Manifest

Authoritative Infrastructure, Architecture Topology, Secrets Register & Maintenance Standard Operating Procedures (SOP)

Target Host: Oracle Cloud (Ampere A1 Flex)	**Production Domain: https://pdsms.online**
Compute Specs: 4 OCPUs (ARM64) • 24 GB RAM	**Public IP:** 129.225.122.78
Database: PostgreSQL 18 (4GB Shared Buffers)	**Reverse Proxy:** Nginx Alpine (Ports 80 & 443 SSL)
Generated: 29 September 2026	**Security Classification: STRICTLY CONFIDENTIAL**

1. System Architecture & Component Boundaries

The CSMS V2 production instance on Oracle Cloud Infrastructure is organized into a strictly isolated, 3-tier containerized stack running on Docker Engine. Public ingress traffic is accepted exclusively by Nginx on ports 80 and 8000. All backend application workers and database processes communicate across a private, internal Docker bridge network (csms_internal_network) with zero outside internet exposure.

Tier	Container Name	Image & Runtime	Exposed Ports / Network Scope	Key Responsibilities
Tier 1 Edge Proxy	`csms_nginx`	`nginx:alpine`	**Host 80:80** **Host 8000:8000**	Rate limiting (anti-brute-force), SSL termination readiness, client buffering, security headers, Gzip compression.
Tier 2 Application	`csms_app`	Python 3.12 (Bookworm) FastAPI / Uvicorn (4 workers)	**Internal:8000** (No host ports)	Business domain logic, transaction postings, inventory ledgers, JWT authentication, PDF reporting.
Tier 3 Database	`csms_db`	PostgreSQL 18 4GB Shared Buffers	**Internal:5432** Loopback 127.0.0.1:5432	Authoritative ACID stock movement ledger, persistent volume csms_v2_pgdata, 14GB RAM ceiling.

2. Asset Manifest: Server-Only vs. GitHub Repository

To safeguard confidential production credentials and maintain operational integrity, repository code is strictly segregated from host-specific production artifacts. The following matrix specifies where every component lives and its confidentiality level:

File / Component Path	Location	In GitHub?	Purpose & Security Classification
`/opt/csms_v2/.env`	**Server Only**	**NEVER**	Contains live PostgreSQL credentials, DB passwords, and cryptographic JWT secret key.
`/opt/csms_v2/docker-compose.prod.yml`	**Server Only**	**NO**	3-tier production orchestration definition tailored for OCI 4 OCPU / 24GB instance.
`/opt/csms_v2/nginx/default.conf`	**Server Only**	**NO**	Nginx reverse proxy with catch-all scanner drop rules (444 / reject handshake), rate limits, and SSL.
`/opt/csms_v2/certbot/conf/`	**Server Only**	**NEVER**	Let's Encrypt live SSL/TLS certificates and cryptographic private keys.

`/opt/csms_v2/certbot/www/`	**Server Only**	**NO**	ACME challenge webroot directory enabling zero-downtime certificate renewals.
`/opt/csms_v2/scripts/backup_db.sh`	**Server Only**	**NO**	Automated nightly database backup script with 14-day retention policy.
`/opt/csms_v2/backups/*.sql.gz`	**Server Only**	**NEVER**	Physical compressed SQL database dumps created daily at 02:00 AM.
`Docker Volume csms_v2_pgdata`	**Server Only**	**NEVER**	PostgreSQL 18 physical cluster mounted to /var/lib/postgresql on NVMe disk.
`/app (Application Source)`	Server & GitHub	**YES**	Core FastAPI Python codebase, web UI routes, Jinja templates, service layer, posting engine.
`/alembic (Migrations)`	Server & GitHub	**YES**	Database schema version history (Migrations 0001 through 0018).
`/scripts (Seed & Maintenance)`	Server & GitHub	**YES**	Base organization, store, role seeds, and diagnostic verification scripts.
`Dockerfile & pyproject.toml`	Server & GitHub	**YES**	Standard container build packaging and Python dependency declarations.
`docker-compose.yml`	Server & GitHub	**YES**	Original single-container baseline preserved for Google Cloud e2-micro compatibility.

3. Production Configuration & Secrets Register

Below is the exact, unredacted production configuration reference active on fastapi-production (129.225.122.78). Use these exact values when performing disaster recovery or server replication.

3.1 Environment File: /opt/csms_v2/.env

```
# ================================================================================
# CSMS V2 - Production Environment (OCI Ampere VM)
# ================================================================================
APP_NAME="CSMS V2"
DEBUG=False

WEB_CONCURRENCY=4

POSTGRES_DB=csms_v2
POSTGRES_USER=csms_user
POSTGRES_PASSWORD=Harshitha@2019

DATABASE_URL="postgresql+asyncpg://csms_user:Harshitha%402019@db:5432/csms_v2"

JWT_SECRET_KEY="62184502c7cbc4290cfdd0b4a580689efa74b14d78efd93b21068d76f08516fa"
JWT_ALGORITHM="HS256"
ACCESS_TOKEN_EXPIRE_MINUTES=480
```

3.2 PostgreSQL 18 Hardware Memory Tuning Parameters

Parameter	Configured Value	Production Purpose & Rationale
`shared_buffers`	**4GB**	Allocates dedicated RAM for caching table pages and inventory indexes.
`effective_cache_size`	**10GB**	Planner advisory indicating total memory available in OS cache + shared buffers.
`work_mem`	**32MB**	Per-sort/hash allocation; protects against OOM spikes under 100 max connections.
`maintenance_work_mem`	**1GB**	Accelerates VACUUM, CREATE INDEX, and large Alembic migration operations.
`shm_size (Docker)`	**1GB**	Expands POSIX /dev/shm; eliminates parallel worker allocation failures.
`random_page_cost`	**1.1**	Optimized for OCI NVMe/SSD block storage; favors fast index lookups.
`effective_io_concurrency`	**200**	Enables deep asynchronous I/O pipelining for bitmap index heap scans.
`statement_timeout`	**60s**	Terminates runaway queries while allowing sufficient window for migrations.
`idle_in_transaction_...`	**60s**	Automatically breaks abandoned sessions holding locks on stock movement tables.
`log_min_duration_statement`	**500ms**	Observability: captures any query taking >500ms in container logs.

4. Standard Maintenance Operating Procedures (SOP)

All operational and maintenance procedures must be performed from the root deployment directory /opt/csms_v2 on the host VM.

SOP-1: Deploying Code Updates from GitHub (Zero DB Downtime)

```
cd /opt/csms_v2

# 1. Pull latest application code from GitHub
git pull origin main

# 2. Execute new Alembic database migrations (if any were committed)
docker compose -f docker-compose.prod.yml run --rm app alembic upgrade head

# 3. Rebuild the application container image using cached Docker layers
docker compose -f docker-compose.prod.yml build app

# 4. Gracefully restart the app container without touching or restarting PostgreSQL
docker compose -f docker-compose.prod.yml up -d --no-deps app
```

SOP-2: Automated and Manual Database Backups

Backups are executed via an automated cron job every night at **02:00 AM** using /opt/csms_v2/scripts/backup_db.sh with a rolling 14-day deletion policy. To trigger an on-demand manual backup before system maintenance:

```
# Trigger immediate manual backup
/opt/csms_v2/scripts/backup_db.sh

# Verify backup creation, file size, and timestamp
ls -lh /opt/csms_v2/backups/
```

SOP-3: Restoring Database from Compressed Backup

```
# Restore compressed database dump into csms_v2 database
gunzip -c /opt/csms_v2/backups/csms_v2_YYYYMMDD_HHMMSS.sql.gz | \
  docker exec -i csms_db psql -U csms_user -d csms_v2
```

SOP-4: System Monitoring, Log Inspection & Health Verification

```
# Check live health status of all 3 containers
docker compose -f docker-compose.prod.yml ps

# Monitor real-time CPU & RAM consumption across containers
docker stats csms_db csms_app csms_nginx

# Follow live application logs with per-request processing times
docker compose -f docker-compose.prod.yml logs -f app

# Follow Nginx access logs and rate-limiting activity
docker compose -f docker-compose.prod.yml logs -f nginx

# Follow PostgreSQL logs (slow queries >500ms, lock alerts)
docker compose -f docker-compose.prod.yml logs -f db
```

SOP-5: Zero-Downtime SSL Certificate Renewal (Webroot)

SSL certificates are renewed automatically every Monday at **03:00 AM** via webroot challenge without stopping Nginx. To simulate and verify a dry-run renewal at any time:

```
# Test simulated zero-downtime renewal against Let's Encrypt staging
docker run --rm \
  -v /opt/csms_v2/certbot/www:/var/www/certbot \
  -v /opt/csms_v2/certbot/conf:/etc/letsencrypt \
  certbot/certbot renew --webroot -w /var/www/certbot --dry-run

# Active crontab schedule:
# 0 3 * * 1 docker run --rm -v /opt/csms_v2/certbot/www:/var/www/certbot -v /opt/csms_v2/certbot/conf:/etc/letsencrypt cer
```

5. Seeded System Accounts & Access Register

All initial system accounts have been hashed using **Argon2id**. The initial default password for all seeded accounts is: **Password@1**. Individual user passwords should be changed upon first login through user management administration.

Username	Full Name / Designation	Role Code	Assigned Store Scope	Operational Permissions & Authority
dev_admin	System Administrator	`SYSTEM_ADMIN`	Directorate Level (Global Masters)	Full organization, office, store, item master administration, inventory policy, user setup.
dev_director	Director / Superintendent	`DIRECTOR`	Directorate	High-level oversight, executive audit reports, directorate policy approval.
dev_deputy	Deputy Superintendent, Stock & Stores	`DEPUTY_SUPDT_STORES`	Central Store (CENTRAL)	Central Store controlling officer; authorizes stock verifications, adjustments, returns, petty purchases.
dev_general_sk	General Storekeeper	`GENERAL_STOREKEEPER`	Central Store (CENTRAL)	Executes stock receipts, issues, manual indents, stock verification counts, ledger reviews.
dev_assistant_sk	Assistant Storekeeper	`ASSISTANT_STOREKEEPER`	Central Store (CENTRAL)	Assists General Storekeeper; stock entry, manual indents, physical counts against Central Store.
dev_press_head	Central Press Branch Head	`BRANCH_HEAD`	Central Press (PRESS-STORE)	Branch office controlling officer; approves branch requisitions, verifications, adjustments.
dev_press_sk	Central Press Storekeeper	`BRANCH_STOREKEEPER`	Central Press (PRESS-STORE)	Executes branch stock transactions, issues, receipts from Central Store, petty purchases.

DOCUMENT CONTROL & HANDOVER VERIFICATION	STATUS: PRODUCTION CERTIFIED
Handover Date: 28 September 2026	**Verified Platform:** Oracle Cloud Infrastructure (Ampere A1 Flex)
Database Engine: PostgreSQL 18.x (Tuned 14GB)	**Application Server:** FastAPI Uvicorn (4 Workers / Nginx)